

# CHIRP — Architecture & Workflow

Cuckoo-inspired Hierarchical Intrusion-Resilient Protocol for VANETs

Eight diagrams reconstructed from the repository at commit **07c9aec**. Every number shown is read from `results/*.json` or produced by re-running a named tool in `src/`. Figures labelled ESTIMATED are extrapolations with their assumptions stated; everything else is measured.

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Figure 1 —

## CHIRP — System Architecture

two traffic classes, two authentication paths; member  $\rightarrow$  CH  $\rightarrow$  RSU is the entire topology (no multi-hop)

TIER 0 offline / one-off TIER 1 infrastructure (RSU / sink) TIER 2 cluster heads elected each round by Cuckoo Search TIER 3 vehicles / OBUs 100 nodes 22 readings/round Trusted Authority (TA) issues pseudonym certificates · holds identity escrow offline registration, once 1.  $V \rightarrow TA \{TagCertReq, V, pid, EMS, nv\}k(V, TA)$  2.  $TA \rightarrow V \{TagCert, pid, EMS, V, nv\}sk(TA) \leftarrow 122$  B cert EMS role attribute provisioned requires  $k(V, TA)$ , installed at commissioning — a vehicle without it cannot obtain an EMS cert RSU (roadside unit / sink) urban: 9 in a grid · highway: 5 in a line verifies aggregate MACs and class-2 signatures · anchors global trust what the RSU can actually authenticate fused packet  $\rightarrow$  the HEAD only (fusion loses provenance) class-2 message  $\rightarrow$  the VEHICLE, end-to-end CLUSTER HEAD (10 % of nodes, re-elected every round) class-0 path — FUSE receive  $n$  readings (22 frames) aggregate  $n+1 \rightarrow$  ONE packet attach aggregate MAC (128 b) energy saving lives here delay = cluster size + 1 slots class-2 path — BYPASS ADMISSION DECISION (see D5) reserved slot:  $1/frame = 22/round$  forward VERBATIM, never fused signature survives to RSU delay = 2 slots, constant 1 fused packet + mac\_bits  $128 \times 22$  frames  $n$  verbatim messages each + sig\_bits 576 (64 B ECDSA + 8 B digest) ordinary vehicle 22 class-0 readings/rd joins NEAREST head within tx\_range watchdogs its head EMS vehicle (5 %) 4 of its 22 readings marked class-2 signs + attaches cert digest holds TA-issued EMS role false-priority attacker asserts class-2 on ALL 22 drops nothing — attacks a RESOURCE, not packets NO EMS role attribute ORPHAN no head in range  $\rightarrow$  direct to RSU delay = 1 slot (fastest!) but pays its own  $d^2 / d^4$  orphan bypasses the cluster entirely TDMA within a cluster · 22 frames per round · collision-free by construction. No channel model: transmissions never collide, so no throughput/jitter claims. Slot exhaustion IS modelled — the schedule is a genuinely finite resource. THE STRUCTURAL CONFLICT this architecture exists to resolve: Fusion is the energy saving, and fusion forces every reading to wait for the whole frame. An ambulance cannot wait. The bypass resolves it — and becomes an attack surface.

Figure 2 — System Architecture

Round Workflow — what happens, who pays, and what it costs

one round = 5 phases; energy figures are urban,  $E_0 = 500$  mJ per node

1 MOBILITY STEP vehicles move for  $\text{round\_duration} = 1$  s highway: lanes + wrap urban: grid + turns

2 CH ELECTION Cuckoo Search: 10 nests  $\times$  20 iters, Lévy  $\rightarrow$  Gaussian pool filtered by trust gate and  $\text{ch\_cooldown} = 3$  3 SETUP PHASE ADV

broadcast (heads) JOIN (members) + schedule OUR ADDITION — base paper charges 0 4 STEADY STATE 22 TDMA frames member TX  $\cdot$  head RX

fuse  $\cdot$  forward  $\cdot$  orphans  $\rightarrow$  priority phase 5 TRUST watchdog reports  $\rightarrow$  Beta evidence + assertion rate next round — trust scores feed back into the

CH election pool and the slot-grant gate Per-round energy by role (urban) — why the design decisions came out as they did role cost / round rounds

to death cluster head (8 members) 88.00 mJ 5.7 orphan (direct to RSU) 25.34 mJ 19.7 ordinary member 11.61 mJ 43.0 one re-election (setup) 0.2833

mJ — Serving as head costs  $311\times$  one re-election. Three design decisions that follow from that table 1. Re-elect every round ( $\text{recluster\_max\_rounds} =$

1). Holding heads to save 0.28 mJ of control traffic burned 440 mJ of a head's 500 mJ budget. 2. Add a cooldown ( $\text{ch\_cooldown} = 3$ ) so the same

node cannot be re-elected immediately — FND is governed by how often the SAME node leads. 3. Charge the setup phase. Without it, re-electing

appears free and the stability trigger cannot be evaluated at all. Diagnosed with `src/diagnose_fnd.py` — not guessed from aggregates. THE

REGRESSION INVARIANT — checked before and after every change to `src/` With  $\text{ems\_frac} = 0$  and  $\text{sig\_bits} = \text{mac\_bits} = 0$ , all 55 fingerprinted

configurations must be BIT-IDENTICAL. Holds structurally: `_priority_phase()` returns immediately when no `TrafficModel` is attached, and the model is

attached by the RUNNER — never constructed inside `Network`, which would consume draws from its RNG and shift every mobility trace. The 55

configurations static (paper reproduction) 7 protocols  $\times$  3 seeds = 21  $\cdot$  highway  $7 \times 2 = 14$   $\cdot$  urban  $7 \times 2 = 14$   $\cdot$  trust under attack 3 fractions  $\times$  2 seeds

= 6 Each fingerprinted with SHA-256 over raw float64 bytes of: the full 100-node energy vector, PDR counters, per-round orphan and intra-distance

traces, trust scores. Not "the means agreed to 4 dp" — the 100-element energy vector hashed to the same 16 hex characters. Command: `python3`

`src/regression_gate.py --diff results/regression_reference.json /tmp/now.json`

Figure 3 — Round Workflow and the Regression Invariant

### The TDMA Frame and the Delay Model

delay is measured in SLOTS, never milliseconds — there is no channel and no time model below the round, so a millisecond figure would be invented

One TDMA frame in a cluster of 8 members (22 such frames per round)

RESERVED class-2 slot M1 M2 M3 M4 M5 M6 M7 M8 FUSE + FORWARD 1 packet → RSU slot 0 member slots 1 ... n head's own slot class-0

AGGREGATED delay = cluster size + 1 = 9 slots the head cannot transmit until EVERY member has spoken, because it is fusing them all — so a reading's own slot position is irrelevant class-2 BYPASSED delay = 2 slots (reserved slot + immediate forward) — CONSTANT, independent of cluster size ORPHAN delay = 1 slot — direct to RSU, nothing to wait for. The FASTEST path in the network. deadline\_slots = 4 THE CONSEQUENCE — coverage and emergency latency are in direct opposition Orphans skip fusion, so they are the low-delay path. A protocol that is GOOD at coverage has few orphans, therefore larger clusters, therefore longer frames, therefore later readings — and it has removed the accidental fast path that orphaning provided. Measured across 7 protocols × 2 scenarios, before any bypass existed: protocol urban orphan % delay dl-miss highway orphan % delay dl-miss HEED 46.84.22 0.273 48.05.24 0.387 LEACH 38.56.910.602 31.28.150.609 CSGD-NET 19.37.700.687 10.110.140.820 LEACH-C / PSO / GA 5.9 – 9.28.6 – 9.20.78 – 0.89 0.2 – 1.110.4 – 10.60.90 – 0.95 CHIRP (ours) 3.19.68 0.874 0.110.23 0.958 Read it as a negative result about our own work HEED strands nearly half the network and therefore has the BEST emergency latency in the comparison. CHIRP's headline Module 1 achievement — an 82.7 % / 98.3 % cut in orphan rate — is a LOSS on this metric. Sharper on the highway: no fast path left at all. And why it motivates the bypass Clustering quality alone will always trade coverage against deadlines. You cannot tune your way out of it. If you want both, you need an EXPLICIT fast path that decouples emergency delay from cluster size — which is exactly what class-2 bypass does. Metric definitions — stated precisely, because they are unusual deadline\_miss counts a DROPPED emergency message as a miss. For an ambulance the distinction between "arrived late" and "never arrived" is not worth drawing, and separating them would let a defence look good on latency while quietly losing the message. Pure latency is reported separately as p95 over DELIVERED readings. The deadline applies to class-2 only. Class-0 telemetry is delay-tolerant by definition — that is why it is aggregatable — so it gets a delay distribution instead.

Figure 4 — TDMA Frame and Delay Model

The Priority Admission Decision — and the attack it must survive

Network.\_priority\_phase(), executed per cluster per round

class-2 DEMAND arrives at the head genuine EMS: 4 msgs/round each · attacker: 22 msgs/round each ADMISSION — the security decision none (undefended) every claim admitted the head has no way to check one, so genuine and false demand share the reservation PROPORTIONALLY @20 % att: dl-miss 0.437 11.26 % of network energy drained trust (behavioural) refuse nodes below trust\_gate STATISTICAL — inherits the trust engine's false positives: an ambulance wrongly distrusted loses its slot @20 % att: dl-miss 0.234 halves the damage — never removes it auth (cryptographic) only TA-issued EMS role admitted not a statistical judgement at all — false demand never enters the reservation, so it is never diluted @ EVERY att fraction: dl-miss 0.000 0.00 % drained ·  $p = 5.1e-06$  ARBITRATION against a FINITE reservation capacity = emergency\_slots × packets\_per\_round =  $1 \times 22 = 22$  served = min(admitted, capacity) share = served / admitted (proportional when oversubscribed) SERVED — bypass fusion delay 2 slots · deadline met head refunds aggregation, then pays tx\_energy( $k + \text{sig\_bits}$ , d\_rsu) each FALLBACK — ordinary service delay = cluster size + 1 · DEADLINE MISSED fused into the class-0 packet exactly as if priority did not exist The attack arithmetic — worked example, one cluster capacity = 22 class-2 messages per cluster per round 1 genuine ambulance asserts 4 1 false-priority attacker asserts 22 ← saturates it alone admitted demand 26 share =  $22 / 26 = 0.846$  → the ambulance gets 3.38 of its 4 slots. It loses 15 % to ONE attacker. With two attackers it loses 54 %. Without authentication the head cannot prefer the genuine claim. The asymmetry — why counting vehicles misleads At 5 % attackers in a 100-node network: 5 ambulances × 4 msgs = 20 genuine assertions 5 attackers × 22 msgs = 110 false assertions Equal populations. 5.5× the demand. Measured consequence: 83 % of the reservation's energy already serves liars at just 5 % attackers — identical in both scenarios, because it is a property of the traffic model, not the road. The asymmetry is not in the NUMBER of attackers — it is in how hard each pushes.

Figure 5 — Priority Admission and the Attack

Trust Engine — three layers, two enforcement paths, and why detection  $\neq$  mitigation

WATCHDOG (private signal) member observes: did my head relay the aggregate to the RSU?  $p=0.85$  observed  $\cdot$  5 % error  $\cdot$  22 obs/round  
ASSERTION RATE (public signal) how often does this node claim class-2 priority? (Module 5) needs no consensus — it is stamped on the message  
THREE-LAYER BETA REPUTATION 1  $\cdot$  DIRECT TRUST Beta( $\alpha, \beta$ ) posterior mean, Laplace-smoothed  $\rightarrow$  unobserved nodes sit at 0.5 catches  
blackhole and greyhole 2  $\cdot$  DECAY + ASYMMETRIC PENALTY a failure counts  $\text{trust\_penalty} = 3$  successes  $\cdot$  evidence ages 0.9 PER OBSERVATION  
 $\text{decay\_unobserved} = \text{False}$  — ageing unobserved nodes would rehabilitate an attacker 3  $\cdot$  CREDIBILITY-WEIGHTED RECOMMENDATIONS reports  
fused at the RSU, weighted by each reporter's historical deviation cuts FPR 12–49 % — but does NOT uniformly improve detection, and we say so  
TRUST SCORE gate at 0.40 LEVER 1 — CH eligibility gate a distrusted node cannot be elected head Module 2: works. PDR 0.934  $\rightarrow$  0.957 @20 %  
Module 5: USELESS ALONE. detection 0.925, outcome 0.452  $\rightarrow$  0.463 — asserting priority doesn't need to be a head LEVER 2 — slot-grant gate  
refuse the reserved slot to distrusted nodes added in Module 5 ( $\text{priority\_trust\_gate}$ ) HALVES the damage: 0.437  $\rightarrow$  0.234 but statistical, so it inherits  
false positives: an ambulance wrongly distrusted loses its slot THE TRANSFERABLE LESSON — detection is not mitigation The first trust defence  
detected false-priority attackers at 0.93–0.96 and changed the outcome NOT AT ALL, because the detector was wired to an enforcement path the  
attack does not traverse. Distrusting a node perfectly and then still granting it a reserved slot accomplishes nothing. A detection rate is not a security  
result until you name the enforcement path. A paper reporting "97 % attack detection" has said nothing about outcomes. AND THE FUNDAMENTAL  
LIMIT — a threshold test on rate has a step, and the attacker sits under it genuine EMS asserts at  $4/22 = 0.18$   $\cdot$  detector flags above 0.50  $\cdot$  greed =  
the fraction of its own readings an attacker asserts on greed urban detect highway detect urban dl-miss (none / trust) urban drain % verdict 0.18  
0.0250.004 0.013 / 0.0114.48 invisible, low damage 0.50 0.0250.008 0.206 / 0.2079.08 SWEET SPOT — invisible, 81 % of max damage 0.75  
0.8920.950 0.348 / 0.20710.45 detected — trust now bites At greed 0.50 detection sits AT OR BELOW the false-positive rate — the detector  
contributes nothing — and trust makes the outcome marginally WORSE (0.206  $\rightarrow$  0.207). Lowering the threshold cannot fix it: at 0.18 the attacker's  
per-node behaviour is IDENTICAL to an ambulance's. The distinguishing fact is not behavioural.

Figure 6 — Trust Engine and Detection vs Mitigation

Formal Verification (Module 3a) — Scyther v1.3.0, --max-runs=4

32 claims verified · 8 failing BY DESIGN · one real bug found that the simulator could not see

register.spdl — 8/8 ✓ 1.  $V \rightarrow TA \ V, \{TagCertReq, V, pid, EMS, nv\}k(V, TA)$  2.  $TA \rightarrow V \ \{TagCert, pid, EMS, V, nv\}sk(TA)$  roles: V, TA · claims: Alive, Weakagree, Niagree, Nisynch  $k(V, TA)$  is the provisioning credential — without it the model would describe a TA issuing EMS certs to anyone, which would make the whole defence vacuous. priority.spdl — THE LOAD-BEARING MODEL 3.  $V \rightarrow C \ V, \{TagCert, pid, EMS, V\}sk(TA), nv, \{TagAssert, V, pid, EMS, nv, C\}sk(V)$  4.  $C \rightarrow V \ \{TagGrant, nv, nc, V, C\}sk(C)$  claim\_C4(C, Nisynch) = Ok, proof of correctness the assertion binds the TARGET HEAD C, so an assertion overheard in one cluster cannot be relayed into another What C4 establishes Nisynch over 3 roles requires EVERY preceding event in EVERY role to have a matching partner, in order. So a head's slot grant implies: • a REAL TA run issued that role to that vehicle • a REAL vehicle run asserted it • all three agree on pid, role, nonce • the nonce rules out replay aggregate.spdl — 12/12 ✓ (three deployed message types) chirp-member  $M \rightarrow C \ \{TagReading, M, reading, nm, C\}k(M, C)$  chirp-aggregate  $C \rightarrow R \ \{TagAggregate, C, agg, nc, R\}k(C, R)$  chirp-verbatim  $V \rightarrow R \ \{TagEmergency, V, pid, EMS, emsg, nv, R\}sk(V)$  chirp-verbatim R.Nisynch: the RSU authenticates the VEHICLE end-to-end, through a relay it does not trust at all. fusion-limit.spdl — 0/2, FAILS BY DESIGN Can the RSU authenticate the MEMBER through a fused packet? R.Niagree and R.Nisynch both FAIL. That failure IS the result. The fused value is fresh at the head, not a recoverable function of the readings. A MAC attests to WHO SENT a value, never to whether it faithfully summarises anything. → a compromised head can fabricate it, undetectably. THE BUG SCYTHYER FOUND — cross-protocol MAC confusion Before (no domain separation) — the two data-path MACs were structurally identical: member reading :  $\{M, reading, nm, C\}k(M, C)$  fused aggregate :  $\{C, agg, nc, R\}k(C, R)$  Attack trace (verbatim from --dot-output): Run#2 Alice as HEAD, chirp-AGGREGATE → RSU Charlie: (Alice,agg,nc, {Alice,agg,nc,Charlie}k(Alice,Charlie)) Run#1 Charlie accepts the SAME bitstring as a MEMBER READING. Failure signature — and why it is diagnostic Alive and Weakagree PASSED; Niagree and Nisynch FAILED. The receiver had the right peer but the WRONG MESSAGE MEANING. Fix + re-verification A distinct constant tag inside every MAC and signature, so the message types occupy disjoint formats. All 12 claims then verify, and the suite verifies identically file-by-file OR concatenated. Why the simulator could never find this In network.py a member reading and a fused aggregate are DIFFERENT PYTHON VARIABLES reached by different code paths. There is no bitstring and no wire format, so the confusion cannot be represented. The simulator cannot express the bug. This is the argument for doing formal work. And the architectural payoff of fusion-limit.spdl Because a class-2 message is never transformed, the vehicle's own signature survives the relay to the RSU. So the bypass gains a SECOND justification with nothing to do with latency: Fusion buys energy and costs provenance. The six failing claims in priority.spdl are NOT a vulnerability — diagnosed by experiment, after two wrong guesses Cause: the TA never learns which cluster head the vehicle will later talk to. Registration happens offline and once; message 1 carries no head identity. Weakagree over three roles asks whether EVERY other role is "running the protocol with" the claimer — and the TA is not running with any head. Confirmed: delete the TA role, keep the exchange identical, and all six verify → preserved as priority-assert.spdl (8/8 ✓). Rejected first: (a) "no adversary action" — the trace did contain a redirect node; (b) "the signature doesn't bind V" — binding V changed nothing.

Figure 7 — Formal Verification and the Bug Found

Results — the three defence regimes, both scenarios

EMS deadline-miss vs attacker fraction · 12 seeds · CHIRP · 5 % EMS · crypto charged · source: results/priority\_defence\_\*.json

URBAN

0.0 0.2 0.4 0.6 0.8 EMS deadline-miss 0510203040 attacker % none 0.642 trust 0.392 auth 0.000 HIGHWAY 0.0 0.2 0.4 0.6 0.8 0510203040 attacker % none — every class-2 claim admitted; genuine and false demand share the reservation proportionally trust — statistical gate auth — TA-issued EMS role attribute Energy externalised onto honest heads % of the ENTIRE network's energy budget spent forwarding falsely-urgent messages attacker %urbanhighwayfalse % 55.052.6783 % 2011.266.0195 % 4013.256.8997 % At just 5 % attackers, 83 % of the reservation's energy already serves liars — identical in both scenarios. Why authorisation is categorically different trustauthorisation infers from observed behaviourverifies an unforgeable credential restrained attacker EVADES itcannot be evaded — ignores behaviour has false positivesstructurally impossible non-zero cost at 0 attackersexactly zero cost needs rounds to gather evidenceone signature verification What the crypto costs on the radio (16 seeds, paired Wilcoxon) configurationurbanhighway + aggregate MAC (128 b)−0.17 % (p .63)+0.26 % (p .004) + bypass, no signature+1.08 % (p .083)+0.52 % (p .002) + per-message sig (576 b)+0.71 % (p .058)−0.00 % (p .94) CUMULATIVE+1.63 % +0.77 % Say "we could not detect a per-message signature cost", NOT "the signature is free". ...and where the cost actually is ECDSA P-256 sign 33.4  $\mu$ s · verify 97.4  $\mu$ s = 2.9× and verify is the direction CHIRP does most. Transmitting the 72 B class-2 overhead @90 m: 0.078 mJ Verifying it @25× OBU slowdown, 0.5 W: 1.22 mJ  $\approx$  15× ESTIMATED, not measured — slowdown and power are assumptions, swept. The base paper charges radio only, so its model is structurally incapable of showing this.

Figure 8 — Results — Three Defence Regimes

## Codebase Map and Information Flow

3,867 lines of Python across 24 files, plus 5 Scyther models · legend: ■ model core ■ protocols ■ security ■ drivers ■ verification

src/config.py — 225 lines one Config dataclass · ~70 fields · NO project imports (a leaf) every default carries a comment saying WHY — including the ones deliberately disabled (w\_survival=0 ...) MODEL CORE network.py · 445 run\_setup\_phase() run\_steady\_state() \_priority\_phase() assign\_members, \_place\_rsus energy.py · 28 tx\_energy · rx\_energy aggregation\_energy Eqs. (9)–(11) verbatim — every protocol charged identically mobility.py · 163 StaticMobility HighwayMobility UrbanGridMobility synthetic — not SUMO metrics.py · 111 link\_expiration\_time cluster\_balance deadline\_miss weighted\_percentile SECURITY traffic.py 128 — EMS, demand, token bucket trust.py 183 — Beta, credibility, observe\_priority attacks.py 113 — 6 behaviours incl. falsepriority PROTOCOLS src/protocols/ csgd\_net.py · 142 levy() Mantegna · Eq. 9 the base paper, faithful subclasses chirp.py · 226 ← OURS overrides ONLY fitness/\_pool/select\_ch inherits the Cuckoo engine byte-for-byte baselines — leach.py 76 · heed.py 65 · swarm.py 130 (PSO + GA) swarm.py's own docstring notes: the Latiff cost ALREADY contains the intra-cluster term Eq. 9 omits — which is why PSO and GA hold more energy than CSGD-NET late in the run. EXPERIMENT DRIVERS run\_baseline.py · 124 Table 4 reproduction, 6 protocols run\_module1.py · 202 CH election · defines SCENARIOS run\_module2.py · 188 trust under attack · --ablate run\_priority.py · 226 --bypass-cost · defence sweep · --sweep-greed results/\*.json (13 files) docs/\*.md (10 files) written by hand from those results VERIFICATION & TOOLING regression\_gate.py · 147 — fingerprints 55 configs, SHA-256 over float64 bytes verify\_table4.py · 153 — the 3-argument refutation of Table 4 crypto\_bench.py · 247 — ECDSA/HMAC sizes + timings + energy sweep verify.py 197 · verify\_fit.py 74 — original audit, frame-count fit diagnose\_fnd.py · 106 — identity/history of the first node to die These are why the project's claims are checkable rather than asserted. FORMAL — formal/ (independent of the simulator) register.spdl · priority.spdl · priority-assert.spdl aggregate.spdl · fusion-limit.spdl run\_scyther.py · 168 — runs all five, compares against the expected set in docs/module3.md, exits non-zero if anything moves Shares only the DESIGN with the simulator, never code — which is exactly why it could find a bug the simulator cannot represent. THE ONE FEEDBACK LOOP crypto\_bench.py → measured sizes → config.py defaults → re-run all Module 5 tables